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# cloud security engineer



## A growing specialisation in the cloud industry

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**C**loud technology, just like other prominent fields of tech such as AI, ML and big data, is growing exponentially year after year. Cloud computing brings limitless computing resources, drastically changing how industries, such as IT, function at their core. As cloud computing grew, three prominent names emerged in this burgeoning field – AWS (Amazon Web Services), Microsoft Azure, and Google Cloud. The growing vendors in this market also drove intense competition. New, emerging technologies such as serverless, containers, microservices, blockchain, and more, all spun from the cloud or closely operated with it.

So, why do we need security on the cloud? And why can't we use traditional network security solutions?

Well, traditional network security only makes sense when all the apps are hosted in the data centre and the users are all on the network. However, this has completely changed due to the inception of the cloud. Users and businesses have tremendous amounts of data backed up on the cloud, so it is integral that their (possibly) sensitive data must be kept safe. A set of policies, controls, procedures and technologies must be developed to protect cloud-based systems, data and infrastructure. This is where cloud security engineers or specialists come in.

### WHAT IS CLOUD SECURITY?

As touched upon above, cloud security is a set of strategies and practices developed to protect data and applications hosted in the cloud. There are a wide variety of attacks that can occur in the cloud, therefore cloud security attempts to encompass numerous kinds of possible attacks. A well-designed cloud security strategy can vastly diminish the risk of cyberattacks.

*Note: cloud security is different from SaaS or SECaaS (Security as a service). The latter refers to the security products that are hosted in the cloud.*

Some of the main cloud security risks include data being exposed or leaked, unauthorised user access, an authorised user having excessive access to data, and malicious attacks such as DDoS attacks or malware infections. An efficient cloud security strategy would attempt to significantly reduce the threat posed by these risks by protecting data, managing user authentication and access to data, and staying operational even during an attack.

### KEY TECHNOLOGIES ASSOCIATED WITH CLOUD SECURITY

A cloud security solution must incorporate certain key technologies that are important in the world of the cloud. It would help any aspiring cloud security engineer to familiarise themselves with these processes. The following technologies are the common ones –

**Encryption:** A method which involves scrambling data around so that only the authorised parties can actually understand the haywire information. Chances are that hackers that find their way into a company's cloud will have a significantly tougher time carrying out a number of malicious acts, such as leak, sell or misuse the data, if it has been encrypted. The hacker will come across the scrambled data which cannot be used unless the decryption key is discovered, which is (almost) always impossible.

In the cloud, data can be encrypted at rest (when stored) as well as when in transit (when it is sent from one place to another). Ideally, the data should be encrypted both times to prevent hackers from intercepting the data and accessing it. Also, data in transit includes both data travelling from the cloud to a user, and data travelling from cloud to cloud (multi-cloud or hybrid cloud environment). So, encryption should be performed during both kinds of transfers.